Andrei Dobrete

Curriculum Vitae

linkedin.com/in/Andy3153
github.com/Andy3153
gitlab.com/Andy3153



Self-taught Systems Engineer and Robotics Engineering Graduate focused on network/server management, Linux, NixOS, Docker and shell/Python automation. Hands-on experience in industrial PLC automation and web development.

Work Experience

07/2025 – 08/2025 **Automation & Electrical Engineering Intern**
IMSAT S.A. | `imsat.ro`

- Implemented electrical and control-system modifications at the Ford Otosan plant in Craiova, commissioning and troubleshooting conveyor systems used for door-accessory assembly.
- Programmed Allen-Bradley PLCs using software like RSLogix 5000 and developed HMI applications (FactoryTalk View Studio for PanelView Plus 1000), also used Connected Components Workbench for supporting controllers.
- Created and revised 3D CAD models (Siemens NX, CATIA V5) to validate component fit and support mechanical–controls integration.

06/2024 – 08/2024 **Full-Stack Web Development Intern**
EVASTAFF | `evastaff.ro`

- Implemented and maintained WordPress and Shopify e-commerce platforms on dedicated hosting using cPanel, including server configuration and deployment.
- Designed and developed complete online stores: storefronts, product listings, shopping cart and checkout flows.
- Automated sales and order-fulfillment processes to streamline the selling lifecycle.
- Performed front- and back-end customizations (PHP, CSS, SQL) and developed custom themes/plugins.
- Integrated third-party services and payment gateways via APIs to connect inventory, shipping and CRM systems.

Certifications

03/2025 – Present **Cisco Certified Network Associate (CCNA)**
Hackademy | `hackademy.ro`

- Completed Semesters 1 & 2, covering Network Fundamentals, Switching, Routing, and Wireless Essentials.

Education

10/2026 – Current **Master in Management of Virtual Industrial Enterprises**
Universitatea Națională de Știință și Tehnologie Politehnica București – Faculty of Industrial Engineering and Robotics | `fiir.upb.ro`

10/2022 – 06/2026 **Bachelor of Engineering (B.Eng.) in Robotics**
Universitatea Națională de Știință și Tehnologie Politehnica București – Faculty of Industrial Engineering and Robotics | `fiir.upb.ro`

- **Thesis:** *"Robotic palletizing cell with two inputs and two outputs integrating an articulated arm industrial robot equipped with a multifunctional effector"*

Publications

05/2023 **Phone Controlled Car** | `github.com/Andy3153/phone-controlled-car`
Submissions:
- Student Scientific Communications Session of the Faculty of Industrial Engineering and Robotics, 2023 (`industrialengineeringjournal.eu/pdf/vol7_2023_4/Art9.pdf`)
- Journal of Industrial Engineering and Robotics, Volume 7, Number 4, 2023 (`industrialengineeringjournal.eu/pdf/vol7_2023_4/vol7_2023_4.pdf`, page 54)

Projects

Present **Systems Engineering & Self-Hosted Infrastructure**
- Design and manage NixOS configurations for multi-machine environments (my personal computers and my home server), ensuring consistency and declarative system management.
- Author and maintain custom Nix derivations and reproducible development environments.
- Build and operate Docker Compose deployments and container images for self-hosted server deployments.
- Implement shell and Python automation scripts to streamline daily system operations and maintenance tasks.
- Curate and version-control application configurations (dotfiles) and LaTeX templates for portability and reproducibility.
- Contribute upstream through pull requests, maintaining forks, and submitting patches.

10/2019 – 10/2022 **Lead Programmer, BroBots (FIRST Tech Challenge RO142 | 19088)**
- Primary developer for the robot control codebase using Java, responsible for implementation, debugging and version control.
- Heavily contributed to mechanical design, assembly and prototyping.
- Contributed to iterative testing and performance tuning for competition readiness.
- Mentored newer team members and facilitated cross-functional collaboration between software and mechanical subteams.

Digital Skills

	○ Linux	○ Docker/Podman	○ System Administration
Infrastructure	○ NixOS	○ systemd	○ QEMU/KVM
	○ Python	○ Git	○ Java
	○ Lua	○ C	○ Ladder Logic
	○ Nix	○ C++	
Development	○ POSIX Shell (bash/sh)	○ Arduino	
	○ Siemens NX	○ AutoCAD	○ FreeCAD
CAD/CAM	○ CATIA V5	○ Blender	
Office	○ LaTeX	○ Microsoft Office	○ LibreOffice

Volunteering

03/2024 – Present **Volunteers Polytechnic Bucharest – Member** | `unstpb.ro`

10/2022 – 06/2023 **The Student Organization of the Faculty of Industrial Engineering and Robotics – Member** | `osfiir.ro`

07/2019 – 12/2022 **National Red Cross Society Râmnicu Vâlcea – Member** | `crucearosievalcea.ro`

Language Skills

Romanian Native

English B2 (Professional Working Proficiency)

Driving License

08/2021 Full EU – Categories **B**, **B1**, **AM**